

The artifact selected for this category is a full stack dashboard application which was developed in CS 340 Advanced Programming Concepts. The application was designed for the fictional company, Grazioso Salvare, which was an international search and rescue training organization. The purpose of the dashboard was to allow for users to filter and visualize data from animal shelters in order to identify potential candidates for their rescue training programs. The application was built through the use of Python, MongoDB and the Dash framework. The system integrates a MongoDB database with a python CRUD module to retrieve and manipulate data. The dashboard then displayed the data within an interactive table and provided a pie chart to visualize breed distribution as well as a map showing the location of selected animals. Users could then filter the data by rescue category, like water or mountain rescue to allow for quick identification of suitable animals for the training programs. The artifact includes the dashboard implementation in Python as well as the CRUD module for communication with the MongoDB database.

As for the justification on why this artifact was selected for my ePortfolio; it would be due to the fact that it demonstrates my overall ability to design, implement and enhance a full-stack software application. Key components that showcase my skills in this would be the following: the inclusion of the CRUD module, which connects to MongoDB, interactive dashboard design, the organization and modularity of the code, organization and maintainability of the code (as well as notes to provide easy documentation throughout), and increased focus on security.

I do believe for the most part that I was able to achieve my planned enhancements. One of the main enhancements being the fact that it was moved from the Jupyter Notebook setup to a more modular python program. The new code structure separates the layout, callbacks, services, as

well as database integration. The enhanced CRUD module includes logging as well as error handling to improve the monitoring capabilities of the system and the troubleshooting experience. The Dash layout and assets folder provide a more maintainable and overall cleaner layout. I do believe that these enhancements successfully meet course outcomes such as the following:

Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices. Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals. Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

I found the process of enhancing and refactoring the artifact to be a valuable learning experience. It took me through the process of modularizing a once compact piece of code into separate concerns. I also applied the skills learned from my secure coding class to increase security standards from what was seen in the original project. The main challenge I ran into when addressing my enhancement plan was refactoring from a Jupyter notebook into a structured Python project while still having the same functionality.